Final Report

Date	5 March 2026
Team ID	MBA-041
Project Name	Heart Disease Analysis using Tableau

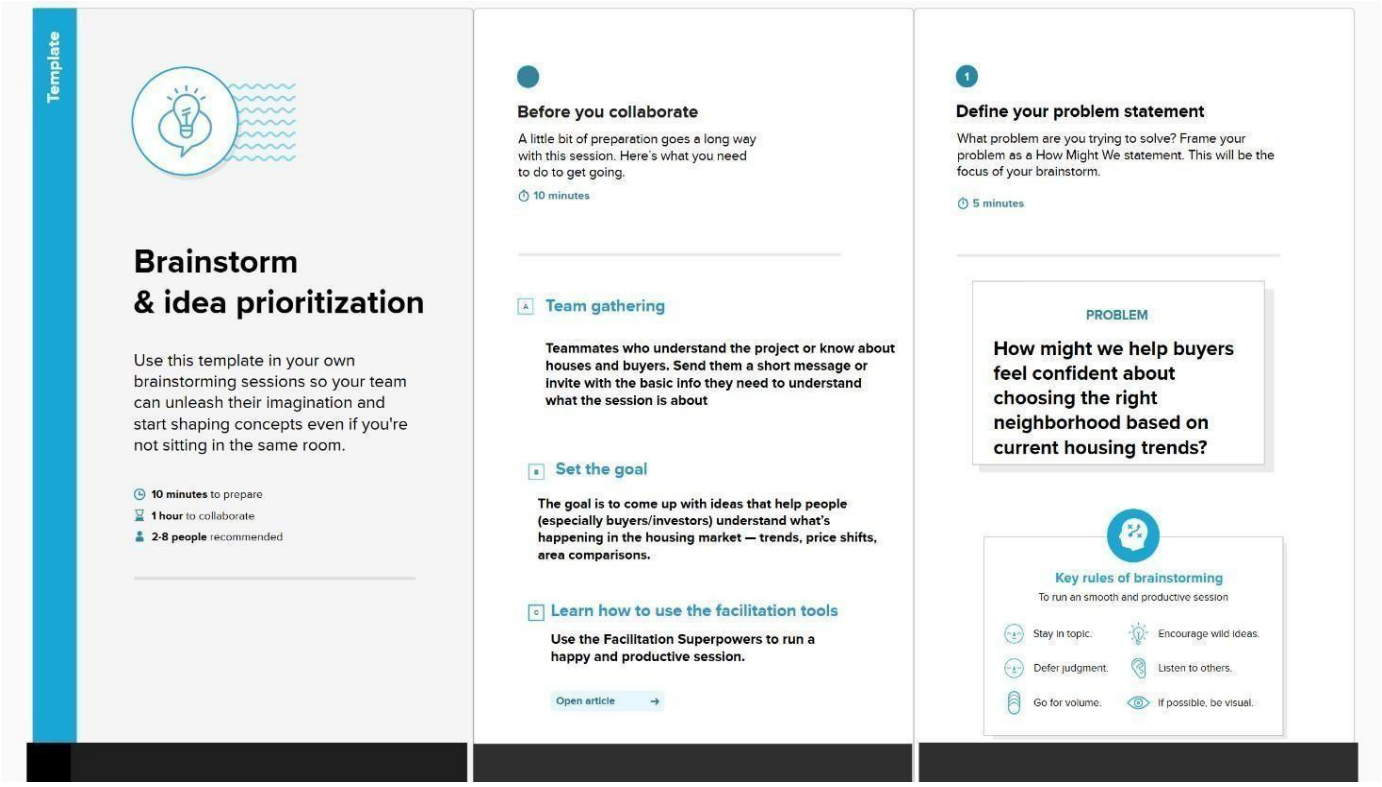
Brainstorm & Idea Prioritization Template:

Brainstorming is a crucial step in the ideation phase, as it helps identify potential solutions, define project direction, and explore various analytical perspectives. During this stage, different ideas were evaluated to determine how data visualization could best address the problem of heart disease risk analysis.

1 Identification of Key Objectives

The brainstorming process began with defining the primary goals of the project:

- Understand the major risk factors contributing to heart disease
- Identify correlations between lifestyle habits and cardiovascular conditions
- Compare disease prevalence across demographic groups
- Provide actionable insights through interactive dashboards
- Enable stakeholders to make data driven decisions

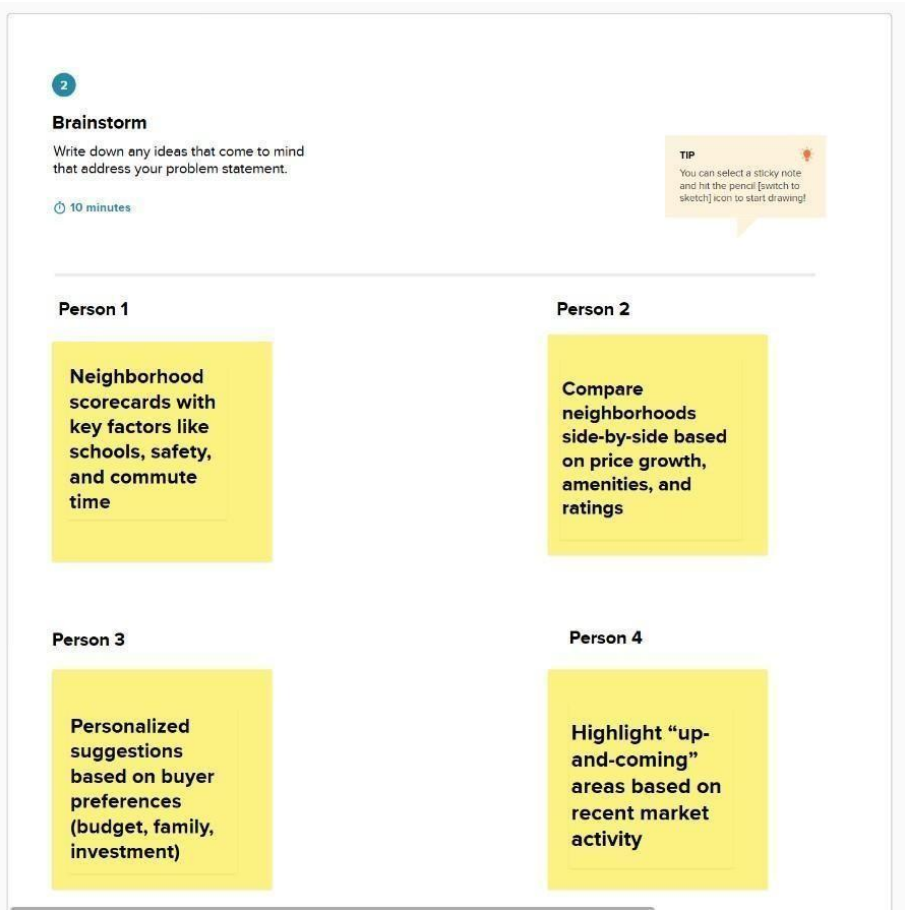


2 Exploration of Data Attributes

Several data attributes were considered important for analysis:

- Demographic Information: Age, Gender, Race
- Lifestyle Factors: Smoking habits, Alcohol consumption, Physical activity
- Medical Indicators: BMI, Diabetes status, Stroke history, General health
- Outcome Variable: Presence or absence of heart disease

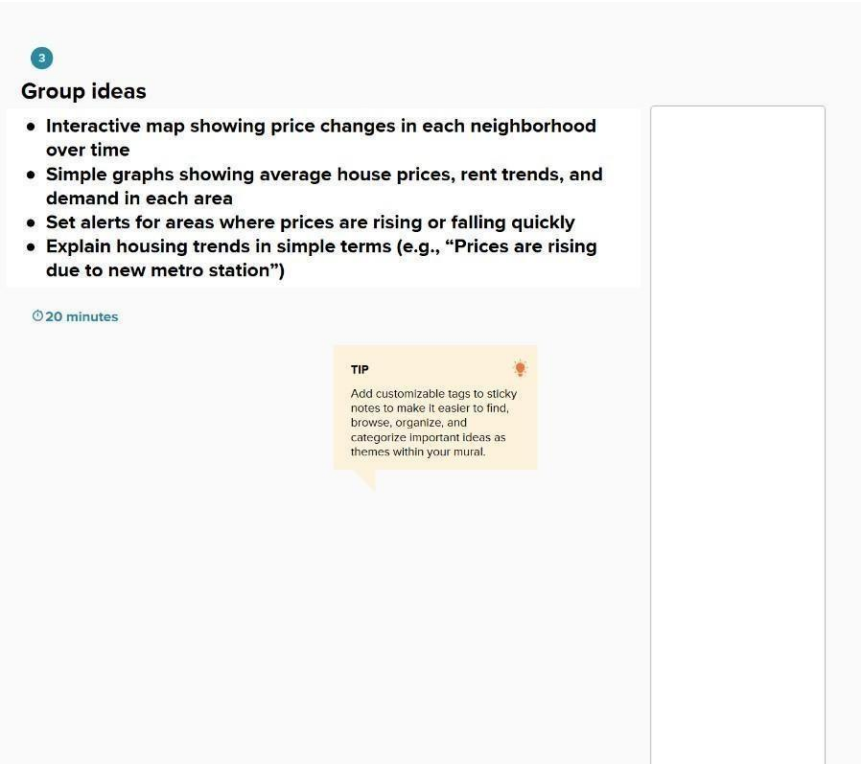
Analyzing these attributes together allows a comprehensive understanding of how multiple factors interact to influence heart disease risk.



3 Visualization Ideas

Different visualization techniques were discussed to effectively represent the data:

- Bar Charts to compare heart disease prevalence across gender and race
- Line or Trend Charts to analyze age-related patterns
- Scatter Plots to explore relationships between BMI and diabetes
- Heat Maps to highlight intensity of risk factors
- Comparative Charts to study the impact of smoking and alcohol
- Dashboards combining multiple visuals for holistic insights



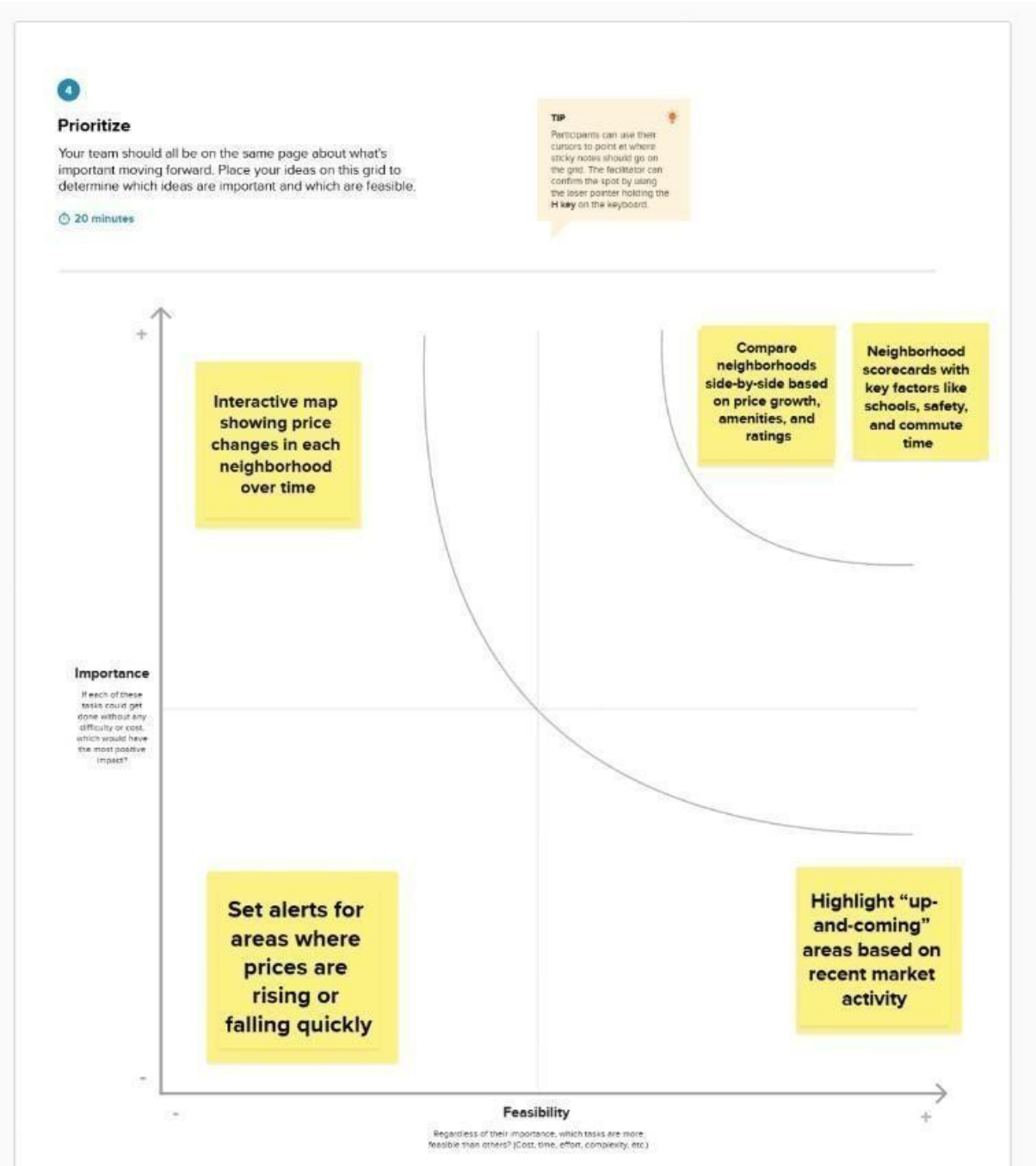
4 Stakeholder Perspective

The brainstorming phase also focused on understanding how different stakeholders would use the insights:

Doctors need patient-level insights to identify high-risk individuals

Government agencies require regional and demographic trends

Individuals benefit from simplified visualizations showing personal risk factors



Ideation Phase Empathize & Discover

Empathy Map Canvas:

An empathy map helps understand the needs, behaviors, and challenges of end users. For this project, the primary users include healthcare professionals, policymakers, and individuals concerned about heart health.

3.1 User Segment 1: Healthcare Professionals (Doctors & Medical Analysts)

Think & Feel

- Concerned about rising heart disease cases

- Need accurate data to support diagnosis and prevention
- Want quick access to patient risk insights

See

- Increasing number of patients with lifestyle-related diseases
- Complex medical reports and large datasets

Say & Do

- Recommend lifestyle changes to patients
- Analyze patient history and clinical indicators

Pain Points

- Difficulty analyzing large datasets quickly
- Lack of visual tools for identifying patterns

Gains

- Quick identification of high-risk patients
- Better treatment planning and preventive care

3.2 User Segment 2: Government & Policymakers

Think & Feel

- Need to reduce healthcare costs and disease burden
- Focused on preventive healthcare strategies

See

- Regional differences in disease prevalence
- Impact of lifestyle and socioeconomic factors

Say & Do

- Design public health programs
- Allocate resources for awareness campaigns

Pain Points

- Lack of clear insights from raw health data
- Difficulty communicating findings to stakeholders

Gains

- Evidence-based policy decisions
- Clear visual reports for presentations

3.3 User Segment 3: Individuals & Patients

Think & Feel

- Concerned about personal health risks
- Want easy-to-understand information

See

- Health reports with complex medical terms
- Lifestyle habits affecting their well-being

Say & Do

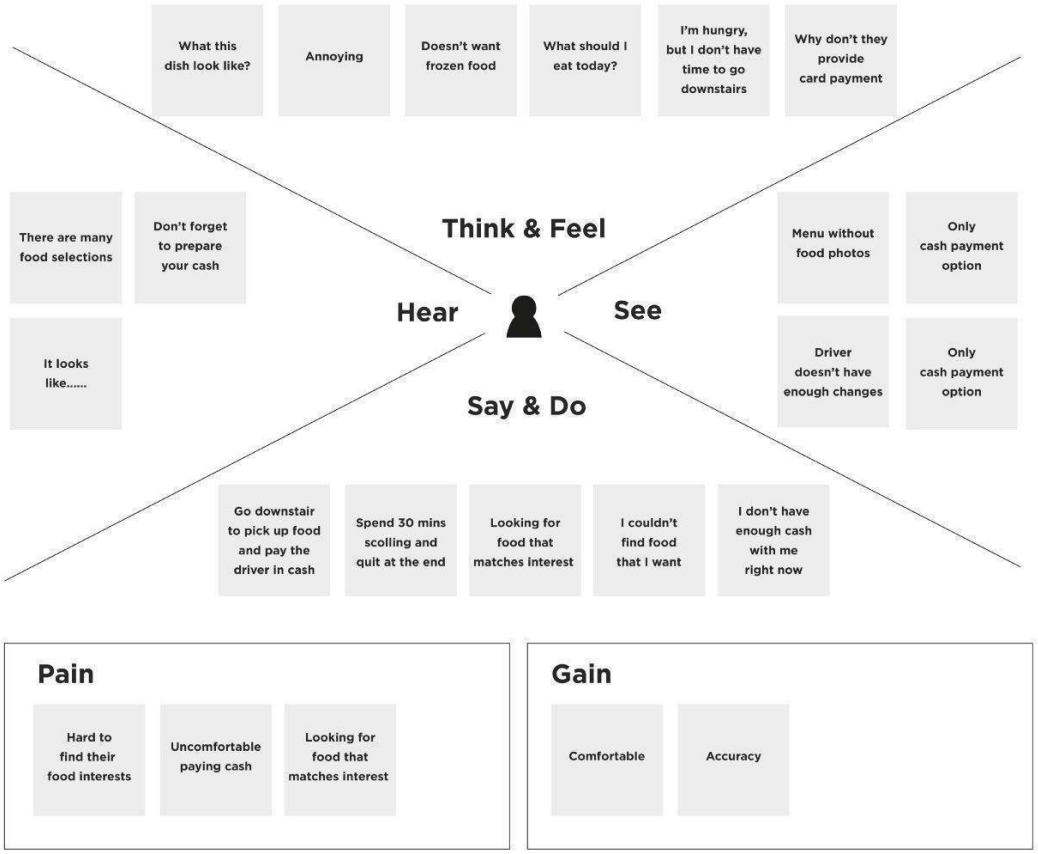
- Seek guidance on improving health
- Monitor personal health metrics

Pain Points

- Difficulty understanding medical data
- Lack of personalized insights

Gains

- Clear understanding of risk factors
- Motivation to adopt healthier habits



Ideation Phase Define the Problem Statements

Problem Statement

Heart disease is one of the leading causes of death worldwide, accounting for millions of fatalities each year. Despite advancements in medical science, the prevalence of cardiovascular diseases continues to rise due to factors such as sedentary lifestyles, unhealthy dietary habits, stress, smoking, alcohol consumption, and lack of physical activity. Early detection and preventive care are crucial in reducing mortality rates and improving overall public health outcomes. However, healthcare organizations and policymakers often struggle to extract meaningful insights from large volumes of health data, which limits their ability to make data-driven decisions.

The core problem addressed in this project is the lack of accessible and interactive tools to analyze and interpret heart disease–related data effectively. Raw medical datasets contain valuable information about patient demographics, lifestyle patterns, and clinical indicators, but without proper visualization and analysis, identifying trends and correlations becomes challenging. Traditional reporting methods are often static and fail to provide real-time insights or allow dynamic exploration of data.

This project proposes the use of Tableau, a powerful data visualization and business intelligence tool, to transform raw heart disease data into interactive dashboards and visual stories. By leveraging Tableau's capabilities, the project aims to identify key risk factors, analyze patterns among different patient groups, and provide actionable insights for healthcare professionals, government agencies, and individuals.

The primary objective is to develop a comprehensive visualization system that highlights relationships between variables such as age, gender, BMI, smoking habits, alcohol consumption, diabetes, physical activity, and general health conditions. These insights can help medical practitioners identify high-risk populations, assist

policymakers in designing preventive health programs, and empower individuals to make informed lifestyle choices.

Ultimately, this project seeks to bridge the gap between complex healthcare data and meaningful decision-making by presenting insights in a clear, intuitive, and visually engaging manner.

Customer Journey Map

The Customer Journey Map illustrates how different users interact with the Heart Disease Analysis system, from identifying their needs to gaining insights through dashboards. It helps in understanding user behavior, expectations, and touchpoints, ensuring the solution is designed to deliver maximum value.

1.1 User Personas

The system primarily serves three categories of users:

- Healthcare Professionals (Doctors, Analysts)
- Government & Policy Makers
- Individuals & Patients

Each persona has different goals, but all rely on data-driven insights to make informed decisions.

1.2 Stages of the Customer Journey

Stage 1: Awareness

At this stage, users recognize the need for data insights.

- Doctors want to analyze patient risk factors
- Policymakers need trends for public health planning
- Individuals want to understand personal risk

User Need: Access to reliable and visual health insights

Stage 2: Exploration

Users begin interacting with dashboards and reports.

- Filtering data by demographics
- Exploring trends and comparisons
- Identifying correlations

Touchpoints: Tableau dashboards, reports, analytics interface

User Need: Easy navigation and intuitive visuals

Stage 3: Insight Generation

Users interpret visualizations to draw conclusions.

- Doctors identify high-risk patients
- Policymakers analyze regional patterns
- Individuals compare their health indicators

User Need: Accurate and actionable insights

Stage 4: Decision Making

Based on insights, users take action.

- Doctors recommend lifestyle changes
- Government designs awareness programs
- Individuals adopt healthier habits

User Need: Clear recommendations supported by data

Stage 5: Outcome & Feedback

Users evaluate the effectiveness of decisions.

- Improved patient care
- Better policy outcomes
- Increased health awareness

User Need: Continuous monitoring through updated dashboards

Project Design Phase-II
Data Flow Diagram & User Stories

2.1 Data Flow Overview

The Data Flow Diagram (DFD) represents how data moves through the system, from collection to

visualization. **Level 0 – Context Diagram** Input:

- Heart disease dataset
- User queries and filters Process:
- Data storage and processing
- Visualization generation Output:
- Dashboards
- Reports and insights

Level 1 – Detailed Flow

1. Data Collection

- Dataset downloaded from source Stored in MySQL database

2. Data Storage

- Stored in MySQL database

3. Data Processing

- SQL queries for filtering and transformation

4. Data Integration

- Tableau connects to database

5. Visualization

- Charts, dashboards, and stories created

6. User Interaction

- Filters and exploration

2.2 User Stories

Healthcare Professional

"As a doctor, I want to view patient risk factors by age and lifestyle so that I can identify high-risk patients quickly."

Policymaker

"As a policymaker, I want to compare heart disease rates across demographics so that I can design targeted health programs." **Individual User**

"As an individual, I want to see how my lifestyle habits compare to healthy benchmarks so that I can take preventive actions."

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Solution requirements define what the system must do and the constraints it must operate within.

3.1 Functional Requirements

The system should:

- Import and store heart disease data
- Allow SQL-based data processing
- Connect Tableau to the database
- Generate interactive visualizations
- Provide filters for dynamic analysis
- Display dashboards and stories
- Enable web integration using Flask

3.2 Non-Functional Requirements

Performance

- System should load dashboards quickly
- Queries should be optimized

Usability

- Interface must be intuitive
- Visualizations should be easy to interpret

Scalability

- System should handle larger datasets

Reliability

- Ensure accurate and consistent data

Security

- Secure database connections

Project Design Phase-II
Technology Stack (Architecture & Stack)

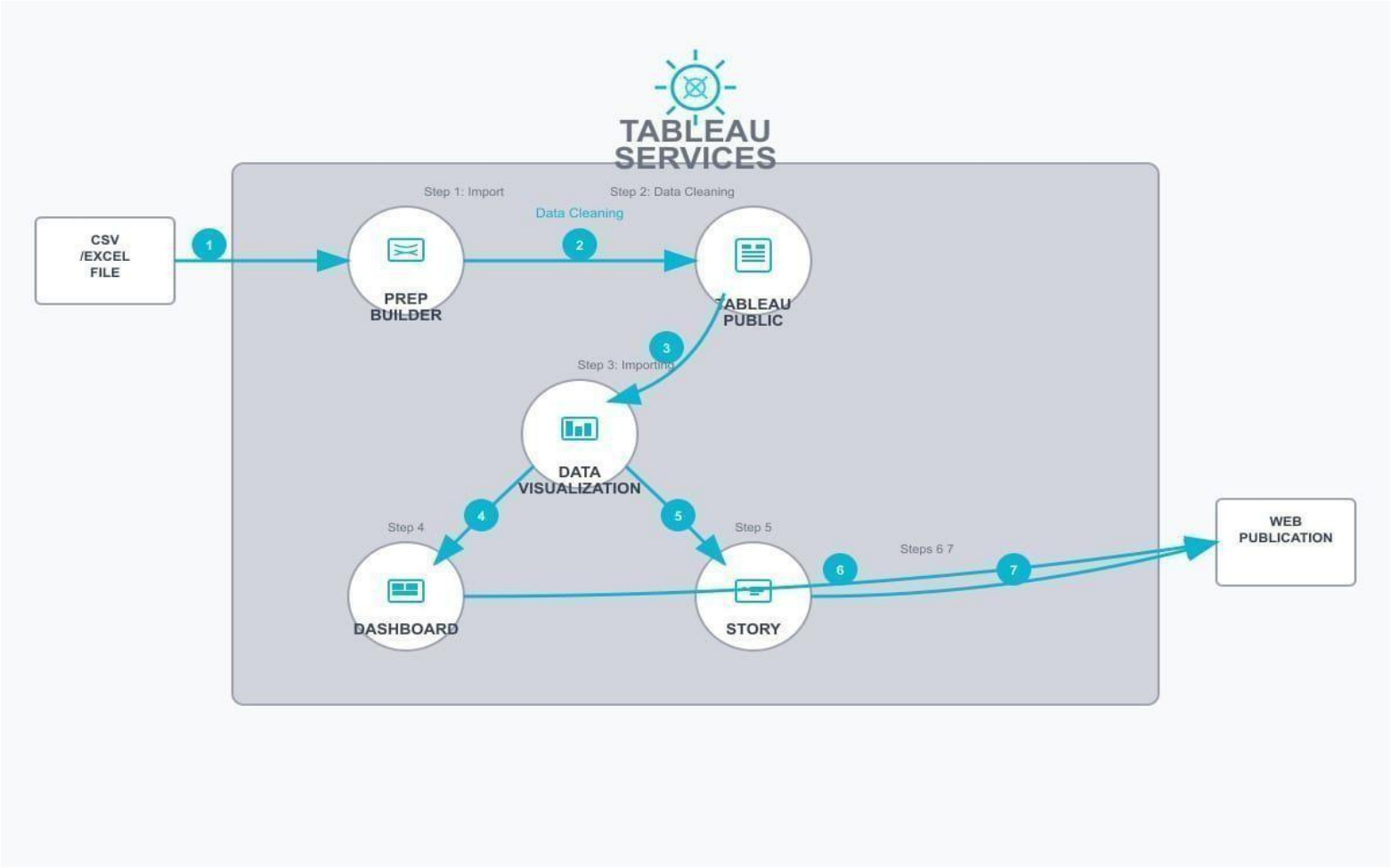
Table-1: Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Web-based dashboards for viewing and interaction	Tableau Public
2.	Application Logic-1	Data preprocessing and transformation workflows	Tableau Prep Builder
3.	Application Logic-2	Interactivity using filters, parameters, and actions	Tableau Filters, Parameters, Actions
4.	Dashboard/Story Logic	Logical flow of insights using story features	Tableau Story Feature
5.	Data Source	Flat files used as housing market datasets	CSV
6.	File Storage	Housing datasets stored locally	Local File System / Google Drive

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	yes	Tableau Public
2.	Security Implementations	N/A	N/A
3.	Scalable Architecture	Can scale by publishing to Tableau Cloud for wider access	Tableau Public
4.	Availability	Dashboards available online 24/7	Tableau Public
5.	Performance	Good \ Better performance	Tableau Public

Technical Architecture:



Project Design Phase Problem – Solution Fit Template

Problem – Solution Fit Template:

The Problem–Solution Fit explains how the proposed system effectively addresses the identified problem and meets user needs. In this project, the primary problem is the difficulty in extracting meaningful insights from large-scale heart disease datasets. Traditional data analysis methods often rely on static reports, which limit the ability to explore data dynamically and identify patterns quickly.

Healthcare professionals, policymakers, and individuals require a solution that simplifies complex medical data and presents it in an intuitive and interactive format. Without proper visualization tools, identifying correlations between lifestyle factors and heart disease risk becomes timeconsuming and inefficient. This gap highlights the need for a data-driven solution capable of transforming raw health data into actionable insights.

The proposed Tableau-based analytics system fits this need by providing interactive dashboards and visual stories that enable users to explore data in real time. Tableau's ability to handle large datasets and generate dynamic visualizations ensures that users can filter, compare, and analyze multiple variables simultaneously. This directly addresses the problem of limited data interpretability.

Additionally, integrating the dataset into a structured database ensures efficient storage and retrieval, while SQL operations allow preprocessing and transformation of data to suit analytical needs. The combination of database management and visualization tools creates a seamless workflow from data acquisition to insight generation.

Overall, the solution aligns strongly with the problem by improving accessibility to insights, enhancing decision-making capabilities, and enabling proactive measures for heart disease prevention. The system not only solves the technical challenge of data analysis but also delivers tangible value to stakeholders by supporting informed actions.

✅ 1. Problem Statement

What is the customer struggling with?

Stakeholders in the real estate sector—such as analysts, marketing teams, and executives—face challenges in:

- Identifying which property features influence pricing trends
- Understanding how renovations affect buyer interest and price
- Making strategic decisions without data-backed insights
- Communicating property trends effectively to clients or investors

💡 2. Existing Behavior / Situation

How are they currently addressing the problem?

- Based on spreadsheets and static reports with limited visual context
- Heavily reliant on manual analysis with potential data misinterpretation
- Missing comprehensive dashboards for interactive exploration
- Not leveraging full potential of historical or feature-specific data

🎯 3. Desired Outcome / Ideal Scenario

What would a better world look like for them?

- Ability to visually explore and interpret housing trends easily
- Understand how each feature (renovation, age, rooms/floors) impacts sale price
- Make informed and fast decisions using an interactive Tableau dashboard
- Identify actionable trends and optimize pricing strategies

🛠️ 4. Our Solution

How does your solution address the problem effectively?

An interactive Tableau dashboard that:

- Provides overall data summaries with key KPIs (Scenario 1)
- Visualizes sales trends based on years since renovation (Scenario 2)
- Breaks down age distribution by renovation status (Scenario 3)
- Analyzes house features like bathrooms, bedrooms, and floors by age (Scenario 4)
- Offers drill-down capabilities and filters for customized insights
- Can be embedded in a web app using Flask for broader accessibility

🚀 5. Why It Works

How does it align with customer behavior and needs?

- Real estate teams already work with data but lack intuitive tools → Tableau adds clarity
- Renovations and house features are top decision factors → directly visualized
- Users prefer visually rich, interactive reports over static spreadsheets
- Immediate insights help close decisions faster, leading to business growth

📣 6. Marketing / Communication Strategy

What messaging and touch-points help adoption?

- Messaging Angle: "Turn your housing data into decisions"
- Emotional Trigger: Solve the frustration of slow, unclear analysis
- Rational Trigger: Improve accuracy and speed of market evaluations
- Touch-points: Demo sessions, tutorials, integration in existing dashboards
- Performance-based dashboards shared with stakeholders

Project Design Phase Proposed Solution Template

Proposed Solution Template:

The proposed solution is an end-to-end data analytics system that leverages Tableau for visualizing heart disease data and generating actionable insights. The system is designed to transform raw healthcare data into interactive dashboards that highlight key trends, correlations, and risk factors. The solution begins with collecting the heart disease dataset and storing it in a relational database such as MySQL. SQL queries are then used to clean, filter, and structure the data to ensure consistency and readiness for analysis. This preprocessing step ensures that the visualizations are accurate and meaningful.

Once the data is prepared, Tableau connects to the database to create a series of visualizations that analyze various aspects of heart disease. These include demographic comparisons, lifestyle impact analysis, and relationships between medical indicators such as BMI, diabetes, and stroke history. Each visualization focuses on a specific analytical objective, and together they provide a comprehensive overview of the dataset.

The dashboards are designed to be interactive, allowing users to apply filters, drill down into specific segments, and explore data from different perspectives. This interactivity enhances user engagement and makes the insights more actionable. For example, doctors can identify high-risk patients, while policymakers can observe trends across populations.

To extend accessibility, the solution also supports web integration using a lightweight framework such as Flask. This enables dashboards and stories to be embedded into a web interface, allowing users to access insights through a browser without needing direct access to Tableau software. The proposed solution is scalable and adaptable, meaning it can be extended with additional datasets or new analytical features in the future. By combining data storage, processing, visualization, and web integration, the system provides a comprehensive platform for heart disease analysis and awareness.

Project Design Phase Solution Architecture

Solution Architecture:

The Solution Architecture describes the structural design of the system, outlining how different components interact to deliver the final analytics solution.

3.1 Data Layer

The data layer is responsible for storing and managing the heart disease dataset. The dataset is stored in a MySQL database, which provides a structured environment for managing large volumes of data. This layer ensures data integrity, consistency, and efficient retrieval.

3.2 Data Processing Layer

The processing layer handles data transformation and preparation. SQL queries are used to:

- Filter irrelevant or redundant data
- Aggregate values for analysis
- Create derived metrics

- Structure data for visualization

This layer ensures that the dataset is optimized for analytical use.

3.3 Analytics & Visualization Layer

This is the core layer where Tableau is used to create visualizations, dashboards, and stories. Tableau connects directly to the database and retrieves processed data for analysis. The visualizations include charts, graphs, and comparative visuals that highlight patterns and correlations.

Key features of this layer include:

- Interactive dashboards
- Real-time filtering
- Drill-down analysis
- Visual storytelling

3.4 Application Layer

The application layer enables web-based access to dashboards using Flask. This layer acts as a bridge between the visualization environment and end users, allowing dashboards to be embedded in a web interface for broader accessibility.

3.5 User Layer

The final layer consists of end users such as healthcare professionals, policymakers, and individuals. Users interact with the system through dashboards and stories, applying filters and exploring insights to support decision-making.

Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Stor
Sprint-1	Data Collection & Extraction	USN-1	**As a user,** I can collect heart disease data from reliable sources, including patient demographics, lifestyle factors, and medical indicators, so that I have a comprehensive dataset for analysis.	2

Sprint-1	Data Preprocessing	USN-2	**As a user,** I can preprocess the dataset to clean and filter unnecessary information such as duplicates, inconsistencies, or irrelevant fields, so that the data is accurate and ready for visualization.	3
Sprint-2	Data Visualization	USN-3	**As a user,** I can visualize heart disease trends using charts, graphs, and comparative visuals, so that I can understand patterns and relationships between risk factors.	3
Sprint-2	Interactive Dashboard	USN-4	**As a user,** I can interact with an interactive dashboard that displays heart disease insights, filters, and performance indicators, so that I can explore data dynamically and gain actionable insights.	2
Sprint-3	User Stories (Dashboard Views)	USN-5	**As a user,** I can use the dashboard to support decision-making, whether for clinical analysis, policy planning, or personal health awareness.	3
Sprint	**Functional Requirement (Epic)**	**User Story Number**	**User Story / Task**	**Stor**
Sprint-3	Web Integration	USN-6	**As a user,** I can integrate the Tableau dashboard into a web interface using Flask, so that the visualizations are accessible online. **As a user,** I can access the dashboard from different devices, so that I can view insights anytime and anywhere.	2

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (a Planned End
Sprint-1	5	2 Days	26 Feb 2026	27 Feb 2026	5
Sprint-2	5	3 Days	28Feb 2026	2 March 2026	5
Sprint-3	5	2 Days	2 March 2026	2 March 2026	5

1. Introduction

Project planning is a critical phase that outlines how the project will be executed, monitored, and completed within the defined timeline and resources. The Heart Disease Analysis project aims to leverage data visualization techniques using Tableau to identify key risk factors and trends associated with
cardiovascular diseases. This phase focuses on defining the project scope, objectives, timeline, resources, and risk management strategies to ensure successful project delivery.

Effective planning ensures that all activities are organized systematically, responsibilities are clearly defined, and potential challenges are identified early. By establishing a structured roadmap, the project team can maintain efficiency, track progress, and ensure alignment with the project goals. **2.**

Project Objectives The primary objectives of the project are:

- To analyze heart disease data using data visualization techniques
- To identify key risk factors such as lifestyle habits and medical indicators
- To create interactive dashboards and visual stories using Tableau
- To support data-driven decision-making for healthcare stakeholders
- To improve awareness and understanding of heart disease trends
-
- These objectives guide the overall direction of the project and define measurable outcomes.

3. Scope of the Project

3.1 In Scope

The project will include:

- Collection and storage of heart disease dataset
- Data preprocessing using SQL
- Visualization and dashboard creation in Tableau

To improve awareness and understanding of heart disease trends

These objectives guide the overall direction of the project and define measurable outcomes.

3. Scope of the Project

3.1 In Scope

The project will include:

- Collection and storage of heart disease dataset
- Data preprocessing using SQL
- Visualization and dashboard creation in Tableau

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-

Story creation to present insights

- Performance testing and optimization

Web integration using Flask

- Documentation and demonstration

3.2 Out of Scope

The project will not include:

- Development of predictive machine learning models
- Real-time patient monitoring systems
- Integration with live hospital databases

Clearly defining the scope helps avoid scope creep and ensures focus on the defined deliverables.

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4. Project Deliverables

The key deliverables of this project include:

Cleaned and structured dataset

- Database setup with SQL queries
- Interactive Tableau dashboards
- Tableau story with analytical insights
- Embedded dashboard using Flask
- Final project documentation

Demonstration video

These deliverables represent the tangible outputs of the project.

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5. Project Timeline

The project will be executed in multiple phases, each with specific tasks and milestones.

Phase 1 Ideation

- Problem definition
- Brainstorming
- Empathy mapping

Phase 2 – Requirement Analysis

- Customer journey mapping
- Data flow diagrams

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• Requirement specification **Phase 3 – Design**

Problem–solution fit
Architecture design

Phase 4 Development

- Data storage and SQL processing

- Tableau visualizations and dashboards

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Phase 5 – Testing

- Performance testing
- Dashboard responsiveness
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Phase 6 Deployment & Documentation • Web

integration Final

documentation

Project demonstration

This phased timeline ensures systematic progress and milestone tracking.

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6. Resource Planning

6.1 Tools & Software

- Tableau Desktop / Tableau Public
- MySQL Workbench
- Python (Flask) • SQL **6.2 Human Resources**

Project Developer (Student)

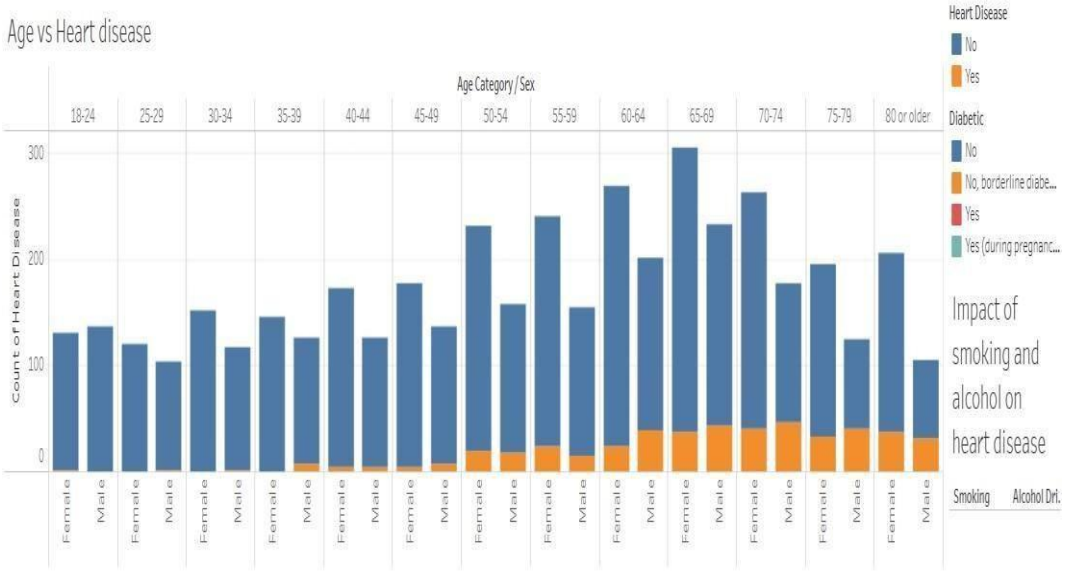
Mentors / Internship Supervisors

6.3 Hardware Requirements

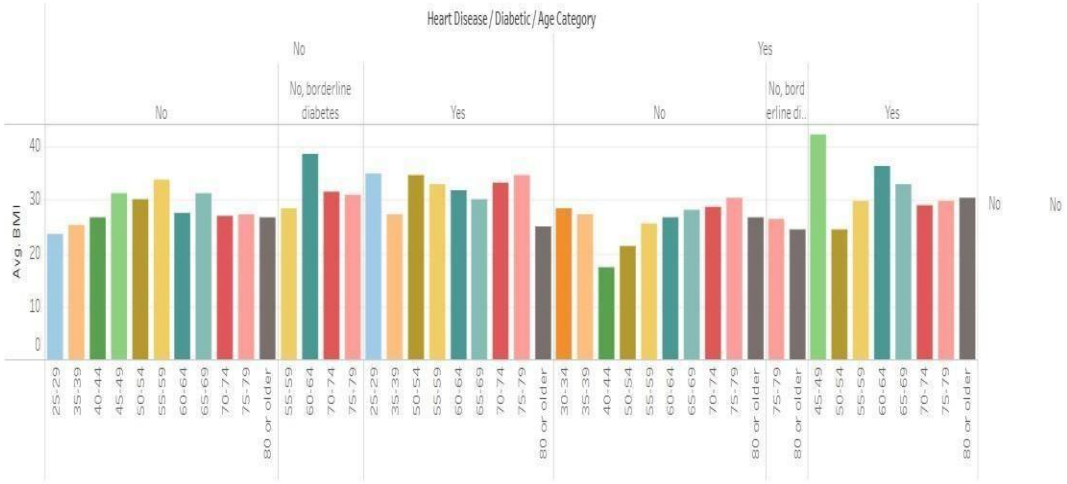
- Computer with minimum 8GB RAM
- Stable internet connection

Proper resource allocation ensures smooth project execution.

Age vs Heart disease

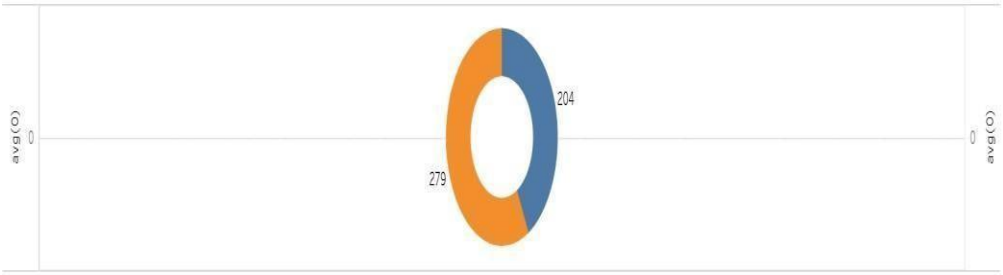


People got stroke suffering from Heart Disease and Diabetic



Race wise Heart Disease

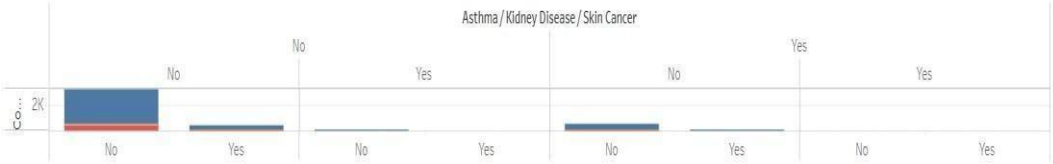
Physical Activity vs Heart Disease



Yes

General Health vs Heart Disease

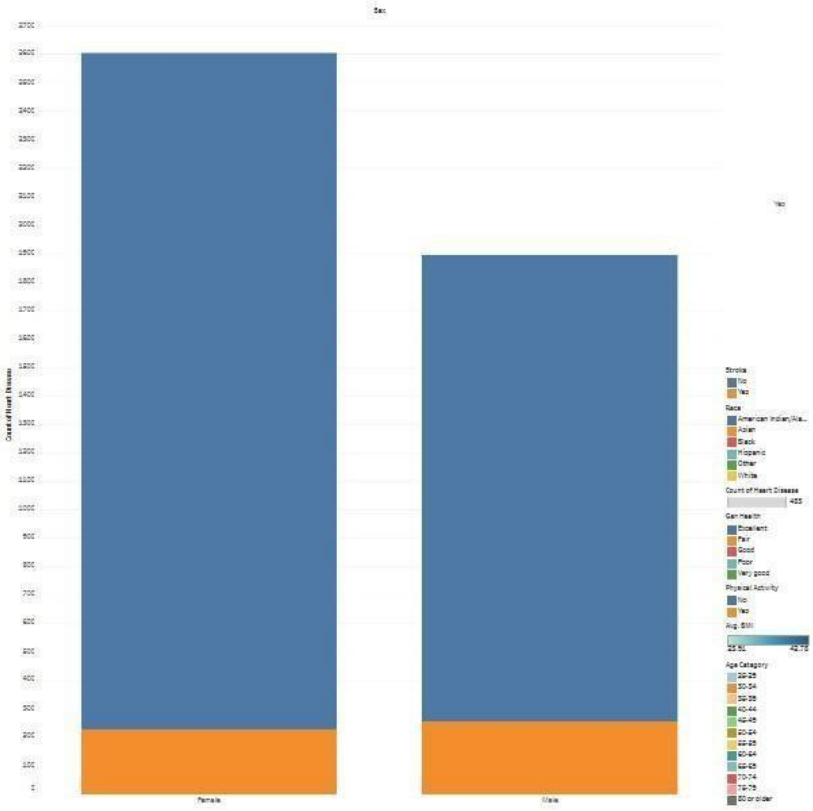
Other Heart Disease vs Stroke



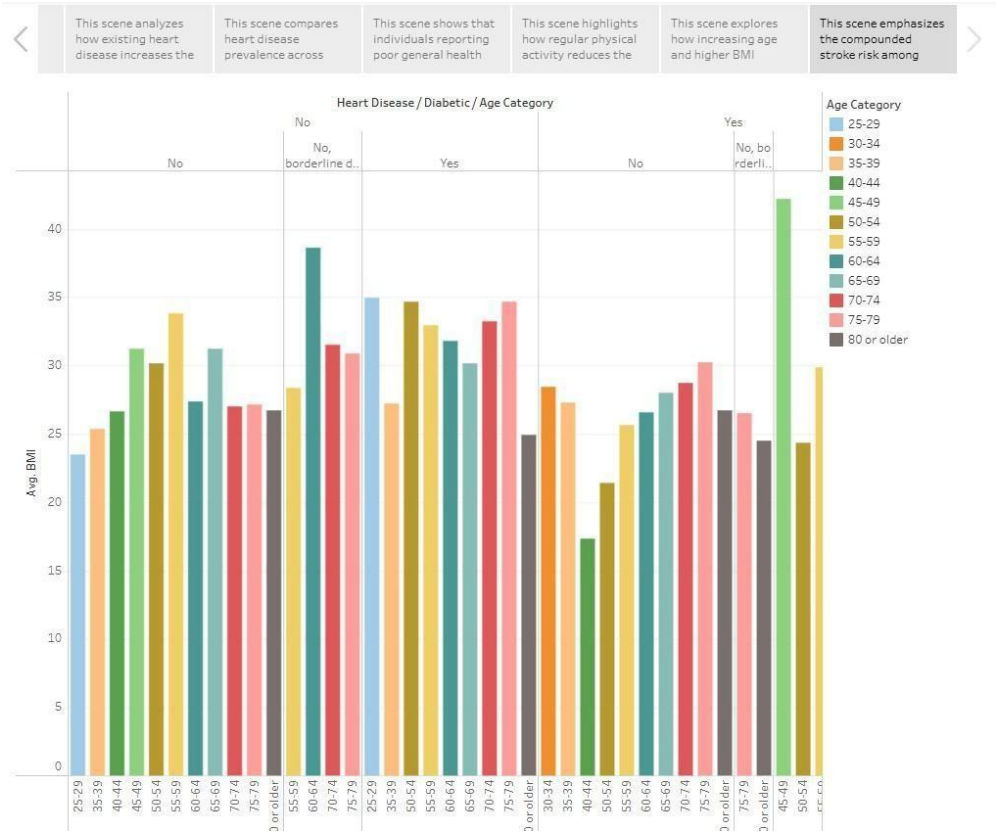
Yes No

Other Heart Disease vs Stroke

Gender vs Heart disease



Story

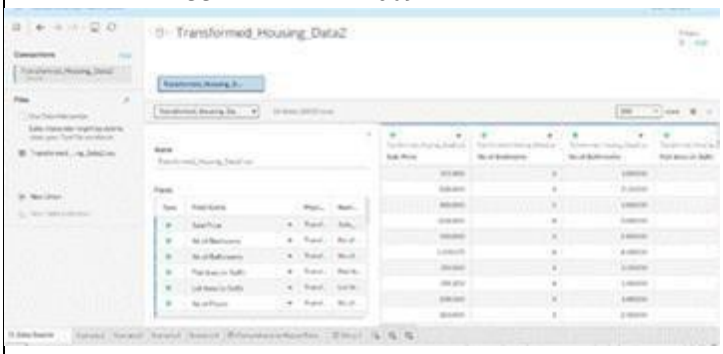
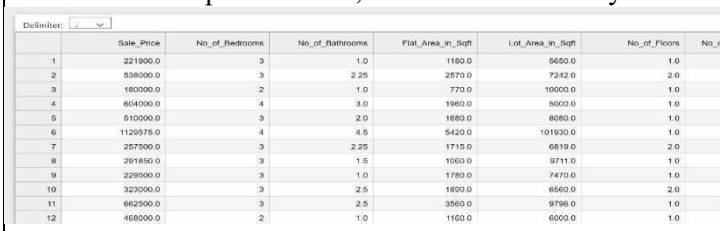

This scene analyzes how existing heart disease increases the
This scene compares heart disease prevalence across
This scene shows that individuals reporting poor general health
This scene highlights how regular physical activity reduces the
This scene explores how increasing age and higher BMI
This scene emphasizes the compounded stroke risk among
Heart Disease / Diabetic / Age Category
No
No, borderline d..
Yes
No
Yes
No
No, bo rderli..
Avg. BMI
0
5
10
15
20
25
30
35
40
Age Category
25-29
30-34
35-39
40-44
45-49
50-54
55-59
60-64
65-69
70-74
75-79
80 or older

Project Development Phase

Model Performance Test

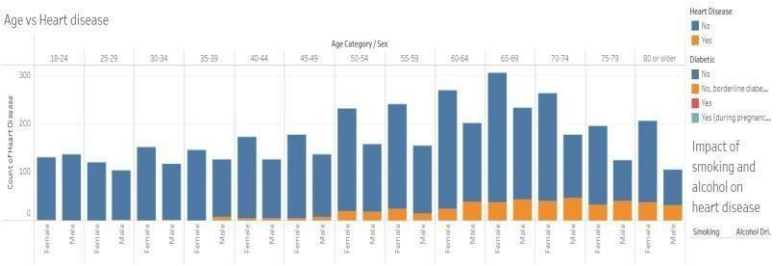
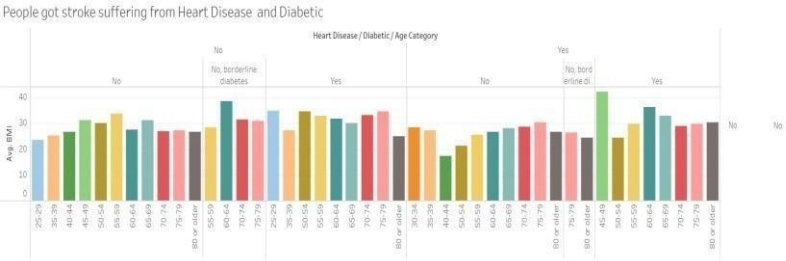
Model Performance Testing:

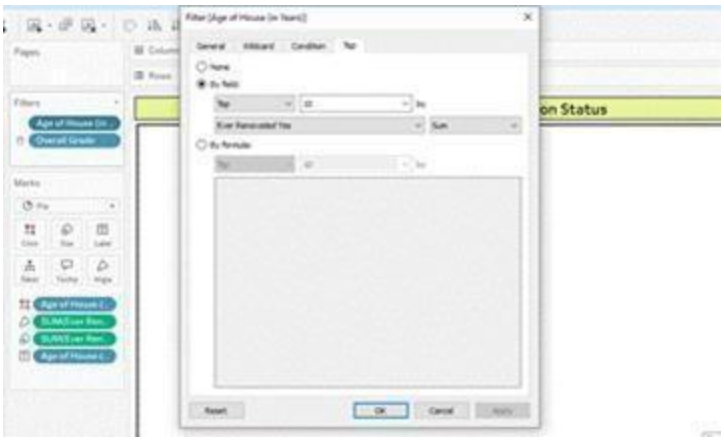
Project team shall fill the following information in model performance testing template.

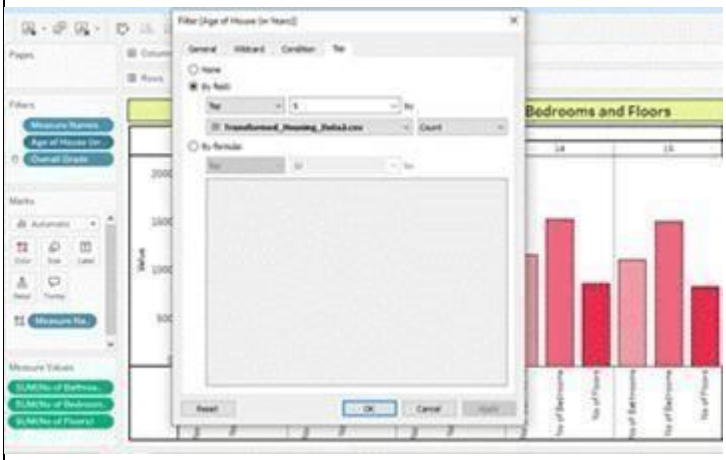
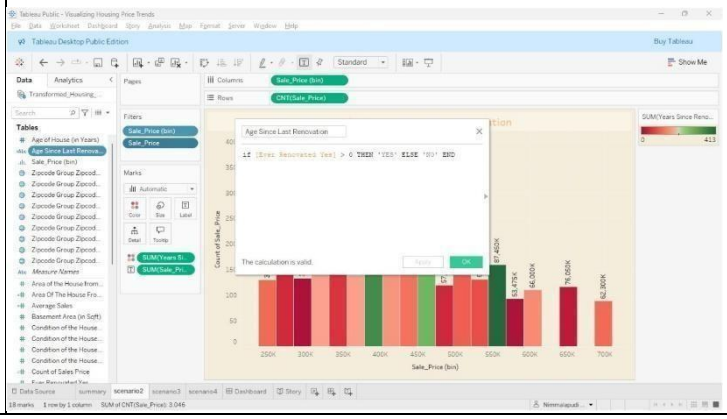
S.No.	Parameter	Screenshot / Values
1.	Data Rendered	Data contains 33 fields and 21609 rows 
2.	Data Preprocessing	Identified 11 duplicate rows , removed unnecessary columns like  zipcodes, no missing or null values.

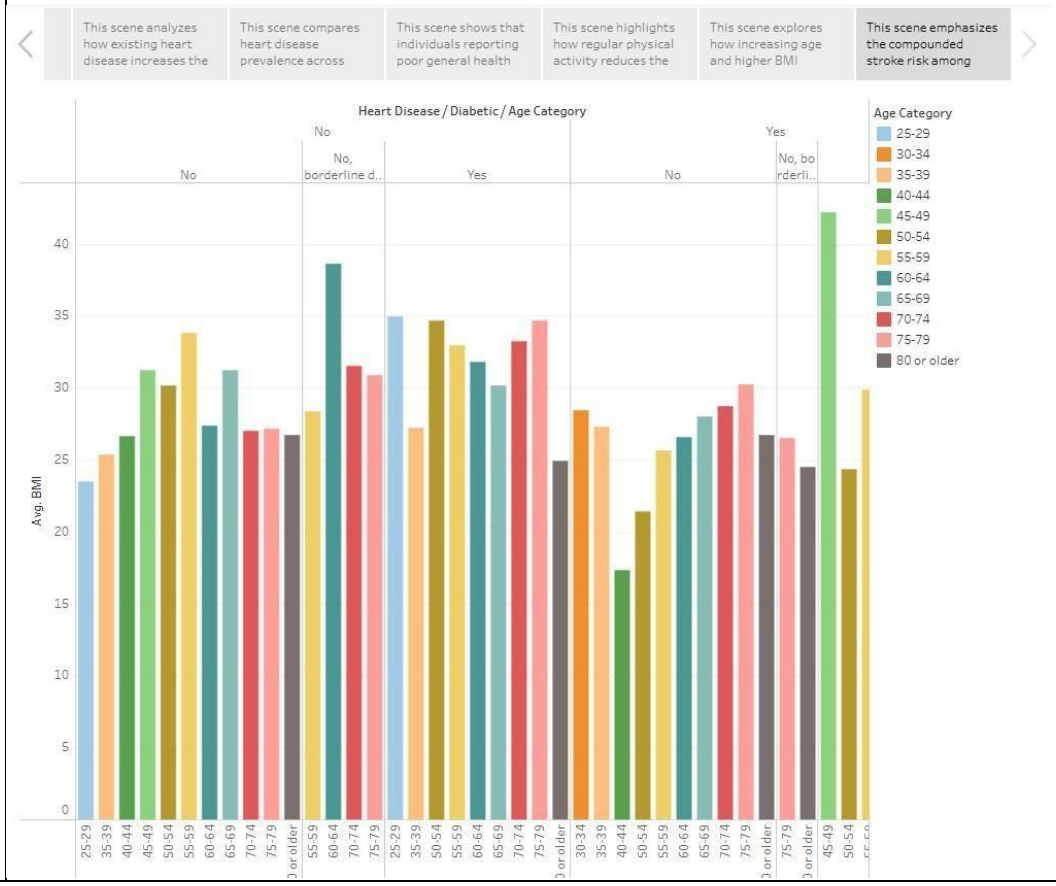
Delimiter: ,

	Sale_Price	No_of_Bedrooms	No_of_Bathrooms	Flat_Area_in_Sqft	Lot_Area_in_Sqft	No_of_Floors	No_
1	221900.0	3	1.0	1180.0	5650.0	1.0	
2	538000.0	3	2.25	2570.0	7242.0	2.0	
3	180000.0	2	1.0	770.0	10000.0	1.0	
4	604000.0	4	3.0	1960.0	5000.0	1.0	
5	510000.0	3	2.0	1680.0	8080.0	1.0	
6	1129575.0	4	4.5	5420.0	101930.0	1.0	
7	257500.0	3	2.25	1715.0	6819.0	2.0	
8	291850.0	3	1.5	1060.0	9711.0	1.0	
9	229500.0	3	1.0	1780.0	7470.0	1.0	
10	323000.0	3	2.5	1890.0	6560.0	2.0	
11	662500.0	3	2.5	3560.0	9796.0	1.0	
12	468000.0	2	1.0	1160.0	6000.0	1.0	

3.	Utilization of Filters	For scenario – 1  For scenario – 2  For scenario – 3

		For scenario – 4 
4.	Calculation fields Used	

5.	Dashboard design	No of Visualizations / Graphs – 4

6	Story Design	No of Visualizations / Graphs – 4